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JAVA LAB EXPERIMENTS

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All Experiment Source Files are available at



<https://github.com/itzspgautam/java-lab>

Lab Experiment 1

Q1. Command Line Arguments

Use command line arguments to perform the following tasks:

- a. Write a message "Hello" on the command prompt using command line arguments.
- b. Use a for loop to display a message of string type for more than one string.
- c. Take two integer type arguments using command line arguments and perform the following operations on the entered input:
 - i. Addition of two numbers
 - ii. Subtraction of two numbers
 - iii. Multiplication of two numbers

Q2. Examination on main() method and static keyword

Write a program to examine the variation in main() method and static keyword in the following cases:

- a. Write a class named Test where it contains a print statement in the static block without the main method. Compile and execute the program. Analyse the program for the compilation and execution.
- b. Add the main() method and perform the compilation and execution of the program. Analyse the difference between the execution of the program in the above two cases.

Q3. Analysis on class name and file name

Create a class named Lab1_Q1 and perform the following:

- a. Create a program where the class name and the file name are the same. Compile and execute the program.
- b. Change the file name so that it does not match with the class name. Compile and execute the program and analyse the error.

Q4. Analysis on public class

Create three classes named Test, Test1, and Test2 and consider the following cases:

- a. Initially, consider all the classes without access modifiers and save with any file name. Compile and execute the program. (Add one class with a main method.)
- b. Add the public keyword in any one class and save it with any file name. After analysing the error, rename the file name to match with the public class name.
- c. Add the public keyword in two of the classes and analyse the compilation and execution phase.
- d. Repeat (b) and (c) by adding main() methods in the corresponding classes.

Lab1_Q1.java

CODE

```
//Lab 1: Question 1
public class Lab1_Q1 {
    public static void main(String[] args) {
        System.out.println("---- (a) Print Hello ----");
        System.out.println("Hello " + args[0]);
        System.out.println("\n---- (b) Display multiple strings ----");
        for (int i = 0; i < args.length; i++) {
            System.out.println("Message: " + args[i]);
        }
        if (args.length ≥ 2) {
            int a = Integer.parseInt(args[0]);
            int b = Integer.parseInt(args[1]);
            System.out.println("\n---- (c) Arithmetic Operations ----");
            System.out.println("Addition: " + (a + b));
            System.out.println("Subtraction: " + (a - b));
            System.out.println("Multiplication: " + (a * b));
        }
        if (args.length ≥ 3) {
            int x = Integer.parseInt(args[0]);
            int y = Integer.parseInt(args[1]);
            int z = Integer.parseInt(args[2]);

            int largest = (x > y) ? ((x > z) ? x : z) : ((y > z) ? y : z);
            System.out.println("\n---- (d) Largest among three ----");
            System.out.println("Largest number: " + largest);

            double avg = (x + y + z) / 3.0;
            System.out.println("\n---- (e) Average of three ----");
            System.out.println("Average: " + avg);
        }
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q1 10 20 30
```

OUTPUT

```
---- (a) Print Hello ----
Hello 10

---- (b) Display multiple strings ----
Message: 10
Message: 20
Message: 30

---- (c) Arithmetic Operations ----
Addition: 30
Subtraction: -10
Multiplication: 200

---- (d) Largest among three ----
Largest number: 30

---- (e) Average of three ----
Average: 20.0
```



Lab1_Q2a.java

CODE

```
//Lab 1: Question 2 (a)
class Lab1_Q2a {
    static void main(String[] args) { //Removed Public
        System.out.println("Hello from main without public!");
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 1 % javac Lab1_Q2a.java
itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q2a
Error: Main method not found in class Lab1_Q2a, please define the main method as:
public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

OUTPUT



Lab1_Q2a.java

CODE

```
//Lab 1: Question 2 (a)
class Lab1_Q2a {
    static void main(String[] args) { //Removed Public
        System.out.println("Hello from main without public!");
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 1 % javac Lab1_Q2a.java
itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q2a
Error: Main method not found in class Lab1_Q2a, please define the main method as:
public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

OUTPUT



Lab1_Q3a.java

CODE

```
//Lab 1: Question 3 (a)
class Test {
    static { //Static block without main() method
        System.out.println("Hello from static block!");
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 1 % javac Lab1_Q3a.java
itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q3a
Error: Could not find or load main class Lab1_Q3a
Caused by: java.lang.ClassNotFoundException: Lab1_Q3a
itzspgautam@Surajs-MacBook-Pro 1 %
```

OUTPUT



Lab1_Q3b.java

CODE

```
//Lab 1: Question 3 (b)
class Lab1_Q3b {
    static {
        System.out.println("Hello from static block!");
    }
    public static void main(String[] args) {
        System.out.println("Hello from main method!");
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q3b
Hello from static block!
Hello from main method!
```

OUTPUT

Lab1_Q4.java

CODE

```
//Lab 1: Question 4
class Test {
    void show() {
        System.out.println("This is class Test");
    }
}

class Test1 {
    void show() {
        System.out.println("This is class Test1");
    }
}

class Test2 {
    void show() {
        System.out.println("This is class Test2");
    }
}

class Lab1_Q4 {
    public static void main(String[] args) {
        Test t = new Test();
        Test1 t1 = new Test1();
        Test2 t2 = new Test2();

        t.show();
        t1.show();
        t2.show();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 1 % javac Lab1_Q4.java
● itzspgautam@Surajs-MacBook-Pro 1 % java Lab1_Q4
This is class Test
This is class Test1
This is class Test2
○ itzspgautam@Surajs-MacBook-Pro 1 %
```

OUTPUT

Lab Experiment 3

1. Suppose, you have rolled an unbiased dice six times consecutively. Write a Java program to display the number of each outcome.

Hints for Solution - The random numbers may be generated by using method `random` of class `Math`.

`import java.lang.Math`

Use `Math.random()` to generate a random number. `random()` method return a value in between 0 to 1]

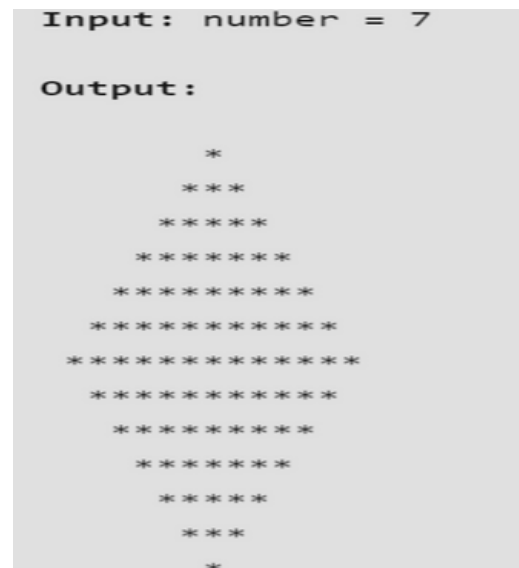
2. Find the largest among 3 user entered numbers at the command prompt in Java.

3. Write a program to calculate area according to user input, whether it is circle, square or triangle (Menu Driven). [Hints: Use switch case]

4. Write a class file – *box* with three data members (length, width, height) and a method *volume()* . Also implement the application *class Demo* where an object of the *box* class is created with user entered dimensions and volume is printed.

Hints: class `Box`

```
{
// create volume ()
{
}
}
Class Demo
{
public static void main(String args[])
{
//create object of class Box
//Take three input through key board
//call volume method
// Print the volume
// Print the three input parameters
}
}
```



The screenshot shows the output of a Java program. At the top, it says "Input: number = 7". Below that, it says "Output:". The output is a diamond shape made of asterisks (*). The pattern is symmetric and has 7 rows of stars. The first row has 1 star, the second has 3, the third has 5, the fourth has 7, the fifth has 5, the sixth has 3, and the seventh has 1. The stars are centered in each row.

5. Write a Java program to print diamond shape star patterns as below.

6. Write a Java program where both static and instance variables are declared, and show their behaviour by creating multiple objects



Lab3_Q1.java

CODE

```
//Lab 3: Question 4
public class Lab3_Q1 {
    public static void main(String[] args) {
        System.out.println("Rolling a dice 6 times...");
        for (int i = 1; i ≤ 6; i++) {
            int outcome = (int)(Math.random() * 6) + 1;
            System.out.println("Roll " + i + ": " + outcome);
        }
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 2 % java Lab3_Q1
Rolling a dice 6 times...
Roll 1: 6
Roll 2: 1
Roll 3: 1
Roll 4: 4
Roll 5: 4
Roll 6: 3
```

OUTPUT



Lab3_Q2.java

CODE

```
import java.util.Scanner;
public class Lab3_Q2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();
        if (a ≥ b && a ≥ c) {
            System.out.println("Largest number: " + a);
        } else if (b ≥ a && b ≥ c) {
            System.out.println("Largest number: " + b);
        } else {
            System.out.println("Largest number: " + c);
        }
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 2 % java Lab3_Q2
Enter three numbers: 10
20
30
Largest number: 30
itzspgautam@Surajs-MacBook-Pro 2 %
```

OUTPUT

Lab3_Q3.java

CODE

```
//Lab 3: Question 3
import java.util.Scanner;
public class Lab3_Q3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("1. Circle");
        System.out.println("2. Square");
        System.out.println("3. Triangle");
        System.out.print("Enter your choice: ");
        int ch = sc.nextInt();

        double area = 0;
        switch (ch) {
            case 1:
                System.out.print("Enter radius: ");
                double r = sc.nextDouble();
                area = 3.14 * r * r;
                break;
            case 2:
                System.out.print("Enter side: ");
                double s = sc.nextDouble();
                area = s * s;
                break;
            case 3:
                System.out.print("Enter base: ");
                double b = sc.nextDouble();
                System.out.print("Enter height: ");
                double h = sc.nextDouble();
                area = 0.5 * b * h;
                break;
            default:
                System.out.println("Invalid choice!");
                return;
        }
        System.out.println("Area = " + area);
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 3 % java Lab3_Q3
1. Circle
2. Square
3. Triangle
Enter your choice: 2
Enter side: 10
Area = 100.0
itzspgautam@Surajs-MacBook-Pro 3 %
```

OUTPUT

Lab3_Q4.java

CODE

```
//Lab 3: Question 4
import java.util.Scanner;
class Box {
    double length, width, height;
    void volume() {
        double vol = length * width * height;
        System.out.println("Volume = " + vol);
    }
}
public class Lab3_Q4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Box b = new Box();

        System.out.print("Enter length: ");
        b.length = sc.nextDouble();

        System.out.print("Enter width: ");
        b.width = sc.nextDouble();

        System.out.print("Enter height: ");
        b.height = sc.nextDouble();

        System.out.println("\nLength = " + b.length);
        System.out.println("Width = " + b.width);
        System.out.println("Height = " + b.height);

        b.volume(); // calling the method
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 3 % java Lab3_Q4
Enter length: 10
Enter width: 20
Enter height: 30

Length = 10.0
Width = 20.0
Height = 30.0
Volume = 6000.0
○ itzspgautam@Surajs-MacBook-Pro 3 %
```

OUTPUT

Lab3_Q5.java

CODE

```
//Lab 3: Question 5
import java.util.Scanner;
public class Lab3_Q5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int n = sc.nextInt();

        // Upper half
        for (int i = 1; i ≤ n; i++) {
            for (int j = i; j < n; j++)
                System.out.print(" ");
            for (int j = 1; j ≤ (2 * i - 1); j++)
                System.out.print("*");
            System.out.println();
        }

        // Lower half
        for (int i = n - 1; i ≥ 1; i--) {
            for (int j = n; j > i; j--)
                System.out.print(" ");
            for (int j = 1; j ≤ (2 * i - 1); j++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 3 % java Lab3_Q5
Enter number: 7

      *
     ***
    *****
   *o*o*o*
  *o*o*o*o*
 *o*o*o*o*o*
*o*o*o*o*o*o*
*o*o*o*o*o*o*o*
 *o*o*o*o*o*o*
  *o*o*o*o*o*
   *o*o*o*o*
    *****
     ***
      *

itzspgautam@Surajs-MacBook-Pro 3 %
```

OUTPUT

Lab3_Q6.java

CODE

```

class ExampleQ6 {
    // static variable (shared by all objects)
    static int count = 0;

    // instance variable (different for each object)
    int number;

    ExampleQ6(int num) {
        number = num; // assign to instance variable
        count++;      // increase static variable
    }

    void showData() {
        System.out.println("Instance variable (number): " + number);
        System.out.println("Static variable (count): " + count);
        System.out.println();
    }
}

public class Lab3_Q6 {
    public static void main(String[] args) {
        ExampleQ6 obj1 = new ExampleQ6(10);
        ExampleQ6 obj2 = new ExampleQ6(20);
        ExampleQ6 obj3 = new ExampleQ6(30);

        obj1.showData();
        obj2.showData();
        obj3.showData();
    }
}

```

```

itzspgautam@Surajs-MacBook-Pro 3 % java Lab3_Q6

```

```

Instance variable (number): 10
Static variable (count): 3

```

```

Instance variable (number): 20
Static variable (count): 3

```

```

Instance variable (number): 30
Static variable (count): 3

```

OUTPUT

Lab Experiment 4

1. Write a program to create a class named Student that have data members as Student_name, Student_roll and Student_fee. Create a parameterized constructor that initialize the data with the values provided at the time of creation of the object. Create another method named Display() to show the values of the two objects. Create the two object and call the Display function.

2. Can we have both Default Constructor and Parameterized Constructor in the same class? Illustrate with a Java program.

3. Program to demonstrate the difference between static and instance variables.

4. Program to demonstrate a static method.

5. Can a Java class contain only a static block without a main() method? Justify your answer with a suitable example.

6. Write a Java program to count the total number of objects created from a class using the 'static' keyword.

7. Implement a Java program, to determine whether a given number is perfect or abundant or deficient.

Perfect

A perfect number is a positive integer that is equal to the sum of its proper divisors(also called its aliquot parts) — that is, all the divisors of the number excluding the number itself. For example:

- 6 is a perfect number because $1 + 2 + 3 = 6$
- 28 is a perfect number because $1 + 2 + 4 + 7 + 14 = 28$

Abundant

A number is abundant when it is less than its aliquot sum. For example:

- 12 is an abundant number because $1 + 2 + 3 + 4 + 6 = 16$
- 24 is an abundant number because $1 + 2 + 3 + 4 + 6 + 8 + 12 = 36$

Deficient

A number is deficient when it is greater than its aliquot sum. For example:

- 8 is a deficient number because $1 + 2 + 4 = 7$
- Prime numbers are deficient

Aliquot sum

Aliquot sum $s(n)$ of a positive integer n is the sum of all proper divisors of n , that is, all divisors of n other than n itself. For example,

aliquot sum $s(6) = 1, 2, 3$

Lab4_Q1.java

CODE

```
//Lab 4: Question 1
class StudentQ1 {
    String student_name;
    int student_roll;
    double student_fee;

    // Parameterized constructor
    StudentQ1(String name, int roll, double fee) {
        student_name = name;
        student_roll = roll;
        student_fee = fee;
    }

    // Method to display student details
    void Display() {
        System.out.println("Name: " + student_name);
        System.out.println("Roll No: " + student_roll);
        System.out.println("Fee: " + student_fee);
        System.out.println();
    }
}

public class Lab4_Q1 {
    public static void main(String[] args) {
        // Creating two student objects with values
        StudentQ1 s1 = new StudentQ1("Amit", 101, 25000.50);
        StudentQ1 s2 = new StudentQ1("Neha", 102, 27000.75);

        // Display their details
        s1.Display();
        s2.Display();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q1
```

```
Name: Amit
Roll No: 101
Fee: 25000.5
```

```
Name: Neha
Roll No: 102
Fee: 27000.75
```

OUTPUT

Lab4_Q2.java

CODE

```
//Lab 4: Question 2
class StudentQ2 {
    String name;
    int roll;
    double fee;
    // Default constructor
    StudentQ2() {
        name = "Unknown";
        roll = 0;
        fee = 0.0;
    }
    // Parameterized constructor
    StudentQ2(String n, int r, double f) {
        name = n;
        roll = r;
        fee = f;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Roll No: " + roll);
        System.out.println("Fee: " + fee);
        System.out.println();
    }
}

public class Lab4_Q2 {
    public static void main(String[] args) {
        // Using default constructor
        StudentQ2 s1 = new StudentQ2();

        // Using parameterized constructor
        StudentQ2 s2 = new StudentQ2("Amit", 101, 25000.50);

        s1.display();
        s2.display();
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q2
Name: Unknown
Roll No: 0
Fee: 0.0

Name: Amit
Roll No: 101
Fee: 25000.5
```

OUTPUT

Lab4_Q3.java

CODE

```

class StudentQ3 {
    // static variable (shared by all objects)
    static String college = "BIT Mesra";

    // instance variable (unique for each object)
    String name;
    int roll;

    StudentQ3(String n, int r) {
        name = n;
        roll = r;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Roll No: " + roll);
        System.out.println("College: " + college);
        System.out.println();
    }
}

public class Lab4_Q3 {
    public static void main(String[] args) {
        StudentQ3 s1 = new StudentQ3("Amit", 101);
        StudentQ3 s2 = new StudentQ3("Neha", 102);

        s1.display();
        s2.display();

        // Changing static variable using one object
        s1.college = "XYZ University";

        System.out.println("After changing college name:\n");

        s1.display();
        s2.display();
    }
}

```

OUTPUT

```

itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q3
Name: Amit
Roll No: 101
College: BIT Mesra

Name: Neha
Roll No: 102
College: BIT Mesra

After changing college name:

Name: Amit
Roll No: 101
College: XYZ University

Name: Neha
Roll No: 102
College: XYZ University

```

Lab4_Q6.java

CODE

```
//Lab 4: Question 6
class Counter {

    static int count = 0;
    Counter() {
        count++; // increase count whenever an object is created
    }

    static void showCount() {
        System.out.println("Total objects created: " + count);
    }
}

public class Lab4_Q6 {
    public static void main(String[] args) {
        Counter c1 = new Counter();
        Counter c2 = new Counter();
        Counter c3 = new Counter();

        Counter.showCount();
    }
}
```

```
public class Lab4_Q5 {
    itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q6
    Total objects created: 3
    itzspgautam@Surajs-MacBook-Pro 4 %
```

OUTPUT

```
itzspgautam@Surajs-MacBook-Pro 4 % javac Lab4_Q5.java
itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q5
Error: Main method not found in class Lab4_Q5, please define the main method as:
    public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```

OUTPUT

Lab4_Q6.java

CODE

```
//Lab 4: Question 6
class Counter {

    static int count = 0;
    Counter() {
        count++; // increase count whenever an object is created
    }

    static void showCount() {
        System.out.println("Total objects created: " + count);
    }
}

public class Lab4_Q6 {
    public static void main(String[] args) {
        Counter c1 = new Counter();
        Counter c2 = new Counter();
        Counter c3 = new Counter();

        Counter.showCount();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q6
Total objects created: 3
○ itzspgautam@Surajs-MacBook-Pro 4 %
```

OUTPUT

Lab4_Q7.java

CODE

```
//Lab 4: Question 7
import java.util.Scanner;
public class Lab4_Q7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive number: ");
        int n = sc.nextInt();
        int sum = 0;

        for (int i = 1; i ≤ n / 2; i++) {
            if (n % i == 0)
                sum += i;
        }
        System.out.println("Sum of proper divisors: " + sum);

        if (sum == n)
            System.out.println(n + " is a Perfect number.");
        else if (sum > n)
            System.out.println(n + " is an Abundant number.");
        else
            System.out.println(n + " is a Deficient number.");
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 4 % java Lab4_Q7
Enter a positive number: 40
Sum of proper divisors: 50
40 is an Abundant number.
○ itzspgautam@Surajs-MacBook-Pro 4 %
```

OUTPUT

Lab Experiment 5

(Access of Inner Class member, Parameter Passing, Method overloading)

EXP 1. Modify the Main class only to print the value of a in the code below:

```
class OuterClass{
    private int a;
    OuterClass(int x){
        a=x;
    }
    class InnerClass{
        void display(){
            System.out.println("a= "+a);
        }
    }
}
```

```
class Main {
    public static void main(String[] args) {
```

```

    _____
    _____
    _____
    }
}
```

EXP 2. Make InnerClass as static (like static class InnerClass) in the code of EXP1, and do the necessary changes to get the value 'a' from display() method.

EXP 3. Write a Java program to demonstrate call by value and call by reference.

EXP 4. Write a program which will overload the area () method and display the area of a circle, rectangle and square as per user choice and user entered dimensions.

EXP 5. Consider a scenario where a bank needs two information of user and need validation. The information are name and age. The name is string type and need to be validated and checked if they are alphabets only whereas the second type in integer and need to be validated if they are greater than 3 and less than 15. Write a program in java where method overloading need to be performed to implement the above task.

[Note: string.matches("[a-zA-Z]+") method is used to determine if a string contains alphabets only]



Lab5_Q1.java

CODE

```
//Lab 5: Question 1
class OuterClass {
    private int a;

    OuterClass(int x) {
        a = x;
    }

    class InnerClass {
        void display() {
            System.out.println("a= " + a);
        }
    }
}

class Lab5_Q1 {
    public static void main(String[] args) {
        OuterClass outer = new OuterClass(10);
        OuterClass.InnerClass inner = outer.new InnerClass();
        inner.display();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 5 % javac Lab5_Q1.java
● itzspgautam@Surajs-MacBook-Pro 5 % java Lab5_Q1
a= 10
○ itzspgautam@Surajs-MacBook-Pro 5 % █
```

OUTPUT



Lab5_Q2.java

CODE

```
class OuterClass {
    private int a;

    OuterClass(int x) {
        a = x;
    }

    static class InnerClass {
        void display(OuterClass obj) {
            System.out.println("a= " + obj.a);
        }
    }
}

class Lab5_Q2 {
    public static void main(String[] args) {
        OuterClass outer = new OuterClass(10);
        OuterClass.InnerClass inner = new OuterClass.InnerClass();
        inner.display(outer);
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 5 % java Lab5_Q2
a= 10
○ itzspgautam@Surajs-MacBook-Pro 5 % █
```

OUTPUT

Lab5_Q4.java

CODE

```
//Lab 5: Question 4
import java.util.Scanner;
class Shape {
    double area(double radius) {
        return Math.PI * radius * radius;
    }
    double area(double length, double breadth) {
        return length * breadth;
    }
    double area(int side) {
        return side * side;
    }
}
public class Lab5_Q4{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Shape shape = new Shape();
        System.out.println("Choose the shape to find area:");
        System.out.println("1. Circle \n2. Rectangle \n3. Square");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:
                System.out.print("Enter radius of circle: ");
                double radius = sc.nextDouble();
                System.out.println("Area of Circle = " + shape.area(radius));
                break;
            case 2:
                System.out.print("Enter length of rectangle: ");
                double length = sc.nextDouble();
                System.out.print("Enter breadth of rectangle: ");
                double breadth = sc.nextDouble();
                System.out.println("Area of Rectangle = " + shape.area(length, breadth));
                break;
            case 3:
                System.out.print("Enter side of square: ");
                int side = sc.nextInt();
                System.out.println("Area of Square = " + shape.area(side));
                break;
            default:
                System.out.println("Invalid choice!");
        }
        sc.close();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 5 % java Lab5_Q4
Choose the shape to find area:
1. Circle
2. Rectangle
3. Square
2
Enter length of rectangle: 10
Enter breadth of rectangle: 5
Area of Rectangle = 50.0
○ itzspgautam@Surajs-MacBook-Pro 5 %
```

OUTPUT

Lab5_Q5_UserValidation.jav

CODE

```
//Lab 5: Question 5
public class Lab5_Q5_UserValidation{
    // Method to validate name - checks if it contains alphabets only
    public boolean validate(String name) {
        // Regex to check if the string contains only alphabets (both uppercase and lowercase)
        if (name != null && name.matches("[a-zA-Z]+")) {
            System.out.println("Name is valid.");
            return true;
        } else {
            System.out.println("Invalid name! Name should contain alphabets only.");
            return false;
        }
    }
    // Overloaded method to validate age - checks if age is between 3 and 15
    public boolean validate(int age) {
        if (age > 3 && age < 15) {
            System.out.println("Age is valid.");
            return true;
        } else {
            System.out.println("Invalid age! Age should be greater than 3 and less than 15.");
            return false;
        }
    }
    public static void main(String[] args) {
        Lab5_Q5_UserValidation uv = new Lab5_Q5_UserValidation();
        uv.validate("Suraj");    // Valid name
        uv.validate("gautam7877"); // Invalid name
        uv.validate(10);        // Valid age
        uv.validate(2);         // Invalid age
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 5 % java Lab5_Q5_UserValidation
Name is valid.
Invalid name! Name should contain alphabets only.
Age is valid.
Invalid age! Age should be greater than 3 and less than 15.
○ itzspgautam@Surajs-MacBook-Pro 5 % █
```

OUTPUT

Lab Experiment 6

EXP 1.- Find the no. of occurrence of each element in an user entered array of numbers.

Int arr [] = {2,34,55,2,34, 6}

EXP 2. Write a Java program to perform the following task:

- (i) create a new array list
- (ii) add 10 elements into it
- (iii) print all the elements
- (iv) remove the third element from the array list
- (v) update 4th array element by another element
- (vi) sort the array list
- (vii) copy the array list into another new array list
- (viii) reverse elements of the new array list
- (ix) join the two array lists
- (x) print the merged array list

Exp 3. Suppose there is bank XYZ. Let us assume that initially, the bank has opened accounts of 5 customers. The customers are having saving account with the initial deposit as given in the following table.

Customer Id Customer name Initial deposit (₹)

1 Maria 200

2 Riya 500

3 Ravi 1000

4 Azmir 500

5 Varun 100

Write a Java program with a class name BankDeposit that displays balance of all the customers along with their names and ID numbers. Using the sorting method, arrange these details in ascending order and display the result.

Let the customer with ID 3 make a transaction and withdraw ₹250 from the account.

Write a Java program that searches for this customer and displays the balance details after this transaction is made.

Exp 4: Write a Java program to return true if an array can be split into two halves in such a way that the sum of left side of the splitting is equal to the sum of the right side.

Sample input 1: The given array is : 1 3 3 8 4 3 2 3 3

Sample output 1: True!! The array can be split in a position where the sum of both side are equal.

Note: Here, splitting position is in between 8 and 4. Sum of left side is $1+3+3+8 = 15$

Sum of right side is $4+3+2+3+3 = 15$

Sample input 2: The given array is : 1 4 1 8 4 3 5 3 3

Sample output 2: False!! The array cannot be split at any position where the sum of both side are equal.

Exp 5: Write a Java program to find the second largest element of an array.



Lab6_Q1.java

CODE

```
public class Lab6_Q1 {
    public static void main(String[] args) {
        int[] arr = {2, 34, 55, 2, 34, 6};
        int length = arr.length;
        boolean[] isVisited = new boolean[length];
        for (int i = 0; i < length; i++) {
            if (isVisited[i] == true) {
                continue;
            }
            int noOfOcc = 1;
            for (int j = i + 1; j < length; j++) {
                if (arr[i] == arr[j]) {
                    noOfOcc++;
                    isVisited[j] = true;
                }
            }
            System.out.println(arr[i] + " occurs " + noOfOcc + " time.");
        }
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 6 % javac Lab6_Q1.java
● itzspgautam@Surajs-MacBook-Pro 6 % java Lab6_Q1
2 occurs 2 time.
34 occurs 2 time.
55 occurs 1 time.
6 occurs 1 time.
○ itzspgautam@Surajs-MacBook-Pro 6 %
```

OUTPUT



Lab6_Q4.java

CODE

```
//Lab 6: Question 4
public class Lab6_Q4 {
    public static void main(String[] args) {
        int[] arr = {1, 3, 3, 8, 4, 3, 2, 3, 3};
        boolean canSplit = false;
        // find total sum
        int totalSum = 0;
        for (int num : arr) {
            totalSum += num;
        }
        int leftSum = 0;
        // check for split position
        for (int i = 0; i < arr.length - 1; i++) {
            leftSum += arr[i];
            int rightSum = totalSum - leftSum;
            if (leftSum == rightSum) {
                canSplit = true;
                System.out.println("True!! The array can be split in a position where the sum of both sides are equal.");
                System.out.println("Splitting position is between index " + i + " and " + (i + 1));
                System.out.println("Sum of left side: " + leftSum);
                System.out.println("Sum of right side: " + rightSum);
                break;
            }
        }
        if (!canSplit) {
            System.out.println("False!! The array cannot be split at any position where the sum of both sides are equal.");
        }
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 6 % java Lab6_Q4
True!! The array can be split in a position where the sum of both sides are equal.
Splitting position is between index 3 and 4
Sum of left side: 15
Sum of right side: 15
```

OUTPUT

```

Lab6_Q2.java

//Lab 6: Question 2
import java.util.ArrayList;
import java.util.Collections;
public class Lab6_Q2 {
    public static void main(String[] args) {

        //i. Creating new arraylist;
        ArrayList<Integer> list = new ArrayList<>();

        //ii. Adding 10 elements into it
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);
        list.add(50);
        list.add(60);
        list.add(70);
        list.add(80);
        list.add(90);
        list.add(100);

        //iii. Printing all elemets
        System.out.println("\n\niii. Experiment 2 List: \n" + list);

        //iv. Removing 3rd Element from list
        System.out.println("\niv. Removed 3rd Element: " + list.remove(2) + "\nNew List: " + list);

        //v.Update 4th element
        System.out.println("\nv. Updating 4th Element: " + list.set(3, 500) + "\nUpdated List: " + list);

        //vi.Sorting list
        System.out.println("\nvi. Existing List: " + list);
        Collections.sort(list);
        System.out.println("Sorted List: " + list);

        //vii.Copying list
        System.out.println("\nvii. Existing List: " + list);
        ArrayList<Integer> list2 = new ArrayList<>(list);
        System.out.println("Copied List: " + list2);

        //viii.Reversing new list
        System.out.println("\nviii. Existing List: " + list2);
        Collections.reverse(list2);
        System.out.println("Reversed List: " + list2);

        //ix. Merging both list
        System.out.println("\nviii. Existing List: " + list + " and " + list2);
        list.addAll(list2);
        System.out.println("Merged List: " + list);

    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 6 % java Lab6_Q2
iii. Experiment 2 List:
[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

iv. Removed 3rd Element: 30
New List: [10, 20, 40, 50, 60, 70, 80, 90, 100]

v. Updating 4th Element: 50
Updated List: [10, 20, 40, 500, 60, 70, 80, 90, 100]

vi. Existing List: [10, 20, 40, 500, 60, 70, 80, 90, 100]
Sorted List: [10, 20, 40, 60, 70, 80, 90, 100, 500]

vii. Existing List: [10, 20, 40, 60, 70, 80, 90, 100, 500]
Copied List: [10, 20, 40, 60, 70, 80, 90, 100, 500]

viii. Existing List: [10, 20, 40, 60, 70, 80, 90, 100, 500]
Reversed List: [500, 100, 90, 80, 70, 60, 40, 20, 10]

viii. Existing List: [10, 20, 40, 60, 70, 80, 90, 100, 500] and [500, 100, 90, 80, 70, 60, 40, 20, 10]
Merged List: [10, 20, 40, 60, 70, 80, 90, 100, 500, 500, 100, 90, 80, 70, 60, 40, 20, 10]

```

OUTPUT

```

//Lab 6: Question 3
import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;
// Class representing a customer's bank deposit
class BankDeposit {
    int id;
    String name;
    int balance;
    // Constructor to initialize customer data
    BankDeposit(int id, String name, int balance) {
        this.id = id;
        this.name = name;
        this.balance = balance;
    }
    // Method to display customer details
    void display() {
        System.out.println("ID: " + id + ", Name: " + name + ", Balance: " + balance);
    }
}

public class Lab6_Q3 {
    public static void main(String[] args) {
        // Create an ArrayList of BankDeposit objects with hardcoded data
        ArrayList<BankDeposit> customers = new ArrayList<>();
        customers.add(new BankDeposit(1, "Maria", 200));
        customers.add(new BankDeposit(2, "Riya", 500));
        customers.add(new BankDeposit(3, "Ravi", 1000));
        customers.add(new BankDeposit(4, "Azmir", 500));
        customers.add(new BankDeposit(5, "Varun", 100));
        // Step 1: Display original list
        System.out.println("Original Customer List:");
        for (BankDeposit customer : customers) {
            customer.display();
        }
        // Step 2: Sort the list by balance in ascending order
        Collections.sort(customers, Comparator.comparingInt(c -> c.balance));
        // Step 3: Display sorted list
        System.out.println("\nSorted Customer List (by Balance Ascending):");
        for (BankDeposit customer : customers) {
            customer.display();
        }
        // Step 4: Search for customer with ID 3 and withdraw 250
        System.out.println("\nAfter 250 withdrawal by Customer ID 3:");
        for (BankDeposit customer : customers) {
            if (customer.id == 3) {
                customer.balance -= 250;
                customer.display();
                break;
            }
        }
    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 6 % java Lab6_Q3
Original Customer List:
ID: 1, Name: Maria, Balance: 200
ID: 2, Name: Riya, Balance: 500
ID: 3, Name: Ravi, Balance: 1000
ID: 4, Name: Azmir, Balance: 500
ID: 5, Name: Varun, Balance: 100

Sorted Customer List (by Balance Ascending):
ID: 5, Name: Varun, Balance: 100
ID: 1, Name: Maria, Balance: 200
ID: 2, Name: Riya, Balance: 500
ID: 4, Name: Azmir, Balance: 500
ID: 3, Name: Ravi, Balance: 1000

After 250 withdrawal by Customer ID 3:
ID: 3, Name: Ravi, Balance: 750
itzspgautam@Surajs-MacBook-Pro 6 %

```

OUTPUT



Lab6_Q5.java

CODE

```
//Lab 6: Question 5
import java.util.Scanner;

public class Lab6_Q5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for (int num : arr) {
            if (num > largest) {
                secondLargest = largest;
                largest = num;
            } else if (num > secondLargest && num < largest) {
                secondLargest = num;
            }
        }

        if (secondLargest == Integer.MIN_VALUE) {
            System.out.println("No second largest element (all elements are equal or only one element).");
        } else {
            System.out.println("Second largest element is: " + secondLargest);
        }
        sc.close();
    }
}
```

```
itzspgautam@Surajs-MacBook-Pro 6 % java Lab6_Q5
Enter number of elements: 4
Enter 4 elements:
10
20
30
40
Second largest element is: 30
itzspgautam@Surajs-MacBook-Pro 6 %
```

OUTPUT

Lab Experiment 7

1. Write a Java program to create a class Student with data members: roll number, name, and CGPA. Use a parameterized constructor to initialize values. Create an array of objects to store the details of n students. Implement a display() method to print the details of all students. Also, customize your code to perform the following task:
 - a. Implement a method to search for a student by roll number and display their name and CGPA.
 - b. Calculate and display the average CGPA of the batch.
 - c. Sort the students in descending order of CGPA and display the rank list with Student information.
 - d. Find and display the student with the highest CGPA and the one with the lowest CGPA

2. Make the following pattern in java using jagged array (create the jagged array that has i column(s) for ith row). The number of rows will be given by user. It may be any integer number. [Jagged array means variable size column in a 2-D array]
0
1 2
3 4 5
6 7 8 9
10 11 12 13 14

3. Write a Java program to return the split point(s) in a 2-D jagged array where negative numbers can be valid inputs. [if an array can be split into two halves in such a way that the sum of left side of the splitting is equal to the sum of the right side.] If no split point is found, return "There is no split point". [Note: There may be multiple split points in an array]

Lab7_Q1.java

CODE

```

//Lab 7: Question 1
import java.util.Scanner;
class Student {
    int roll;
    String name;
    double cgpa;
    Student(int r, String n, double c) {
        roll = r;
        name = n;
        cgpa = c;
    }
    void display() {
        System.out.println("Roll: " + roll + "   Name: " + name + "   CGPA: " + cgpa);
    }
}

public class Lab7_Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();
        Student s[] = new Student[n];
        double total = 0;
        // Input details
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter details of student " + (i + 1) + " :");
            System.out.print("Roll: ");
            int r = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            String nm = sc.nextLine();
            System.out.print("CGPA: ");
            double c = sc.nextDouble();

            s[i] = new Student(r, nm, c);
            total += c;
        }
        // Display all students
        System.out.println("\n--- Student Details ---");
        for (Student st : s)
            st.display();
        // (a) Search by roll number
        System.out.print("\nEnter roll number to search: ");
        int roll = sc.nextInt();
        boolean found = false;
        for (Student st : s) {
            if (st.roll == roll) {
                System.out.println("Found! Name: " + st.name + ", CGPA: " + st.cgpa);
                found = true;
                break;
            }
        }
        if (!found) System.out.println("Student not found!");
        // (b) Average CGPA
        System.out.println("\nAverage CGPA: " + (total / n));
        // (c) Sort by CGPA (descending)
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                if (s[i].cgpa < s[j].cgpa) {
                    Student temp = s[i];
                    s[i] = s[j];
                    s[j] = temp;
                }
            }
        }
        System.out.println("\n--- Rank List ---");
        for (int i = 0; i < n; i++) {
            System.out.println("Rank " + (i + 1) + ": " + s[i].name + " (" + s[i].cgpa + ")");
        }
        // (d) Highest and Lowest CGPA
        System.out.println("\nHighest CGPA: " + s[0].name + " (" + s[0].cgpa + ")");
        System.out.println("Lowest CGPA: " + s[n - 1].name + " (" + s[n - 1].cgpa + ")");

        sc.close();
    }
}

```

OUTPUT

```
itzspgautam@Surajs-MacBook-Pro 7 % java Lab7_Q1
Enter number of students: 4

Enter details of student 1:
Roll: 01
Name: Suraj
CGPA: 8.8

Enter details of student 2:
Roll: 02
Name: Prakash
CGPA: 8.9

Enter details of student 3:
Roll: 03
Name: Gautam
CGPA: 9.0

Enter details of student 4:
Roll: 04
Name: Golden
CGPA: 7.6

--- Student Details ---
Roll: 1 Name: Suraj CGPA: 8.8
Roll: 2 Name: Prakash CGPA: 8.9
Roll: 3 Name: Gautam CGPA: 9.0
Roll: 4 Name: Golden CGPA: 7.6

Enter roll number to search: 2
Found! Name: Prakash, CGPA: 8.9

Average CGPA: 8.575000000000001

--- Rank List ---
Rank 1: Gautam (9.0)
Rank 2: Prakash (8.9)
Rank 3: Suraj (8.8)
Rank 4: Golden (7.6)

Highest CGPA: Gautam (9.0)
Lowest CGPA: Golden (7.6)
```



Lab7_Q2.java

CODE

```
//Lab 7: Question 2
import java.util.Scanner;
public class Lab7_Q2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        int[][] arr = new int[rows][];
        int num = 0;
        for (int i = 0; i < rows; i++) {
            arr[i] = new int[i + 1]; // create ith row with i+1 columns
            for (int j = 0; j ≤ i; j++) {
                arr[i][j] = num++;
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 7 % javac Lab7_Q2.java
● itzspgautam@Surajs-MacBook-Pro 7 % java Lab7_Q2
Enter number of rows: 5
0
1 2
3 4 5
6 7 8 9
10 11 12 13 14
○ itzspgautam@Surajs-MacBook-Pro 7 %
```

OUTPUT

Lab7_Q3.java

CODE

```

import java.util.Arrays;
import java.util.Scanner;

public class Lab7_Q3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input number of rows
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        int[][] arr = new int[rows][];
        int totalLength = 0;

        // Input jagged array
        for (int i = 0; i < rows; i++) {
            System.out.print("Enter number of columns in row " + (i + 1) + ": ");
            int cols = sc.nextInt();
            arr[i] = new int[cols];
            totalLength += cols;

            System.out.println("Enter " + cols + " elements:");
            for (int j = 0; j < cols; j++) {
                arr[i][j] = sc.nextInt();
            }
        }

        // Flatten 2D jagged array into 1D
        int[] flat = new int[totalLength];
        int idx = 0;
        for (int[] row : arr)
            for (int num : row)
                flat[idx++] = num;

        // Find split points
        int totalSum = Arrays.stream(flat).sum();
        int leftSum = 0;
        boolean found = false;

        System.out.println("\nSplit point(s):");
        for (int i = 0; i < flat.length - 1; i++) {
            leftSum += flat[i];
            int rightSum = totalSum - leftSum;

            if (leftSum == rightSum) {
                found = true;
                System.out.println("Between index " + i + " and " + (i + 1)
                    + " (Left = " + leftSum + ", Right = " + rightSum + ")");
            }
        }

        if (!found)
            System.out.println("There is no split point.");

        sc.close();
    }
}

```

OUTPUT

```

itzspgautam@Surajs-MacBook-Pro 7 % java Lab7_Q3
Enter number of rows: 3
Enter number of columns in row 1: 3
Enter 3 elements:
1
3
3
Enter number of columns in row 2: 2
Enter 2 elements:
8
4
Enter number of columns in row 3: 4
Enter 4 elements:
3
2
3
3

Split point(s):
Between index 3 and 4 (Left = 15, Right = 15)
itzspgautam@Surajs-MacBook-Pro 7 %

```

Lab Experiment 8

EXP 1.

For an array, inversion count indicates how far (or close) the array is from being sorted. If array is already sorted then the inversion count is 0. If an array is sorted in the reverse order then the inversion count is the maximum. Formally, two elements $a[i]$ and $a[j]$ form an inversion if $a[i] > a[j]$ and $i < j$. for example if the input is $N = 5$, $arr[] = \{2, 4, 1, 3, 5\}$ then the output will be 3. Since the sequence 2, 4, 1, 3, 5 has three inversions (2, 1), (4, 1), (4, 3)

EXP 2.

Write a program to create a class named Student that have data members as Student_name, Student_roll and Student_fee. Create a parameterized constructor that initialize the data with the values provided at the time of creation of the object. Create another method named Display() to show the values of the two objects using this keyword. Create the two object and call the Display function.

EXP 3. Consider the EXP 2 and create one non parameterized constructor. Create the two objects where one object takes no argument and another object takes arguments to initialize the object. The non parameterized constructor calls the parameterized constructor using this keyword to give a glimpse into constructor chaining. Create another method named Show() that takes one argument of type object. Call the method Display() from the method show() where this is passed as an argument. Now display the object values.

```
//Lab 8: Question 1
import java.util.Scanner;
public class Lab8_Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of the array: ");
        int arraySize = sc.nextInt();
//Input array
        int[] arr = new int[arraySize];
        for (int i = 0; i < arraySize; i++) {
            System.out.print("Enter " + (i + 1) + " Element of array: ");
            arr[i] = sc.nextInt();
        }
//displaying array
        for (int i = 0; i < arraySize; i++) {
            System.out.print(arr[i] + " ");
        }
        int count = 0;
        for (int i = 0; i < arraySize; i++) {
            for (int j = i + 1; j < arraySize; j++) {
                if (arr[i] > arr[j]) {
                    count++;
                }
            }
        }
        System.out.println("\nTotal number of inversion: " + count);
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 8 % java Lab8_Q1
Enter size of the array: 5
Enter 1 Element of array: 2
Enter 2 Element of array: 4
Enter 3 Element of array: 1
Enter 4 Element of array: 3
Enter 5 Element of array: 5
2 4 1 3 5
Total number of inversion: 3
○ itzspgautam@Surajs-MacBook-Pro 8 %
```

```
//Lab 8: Question 2

class Student {
    String Student_name;
    int Student_roll;
    double Student_fee;
    Student(String Student_name, int Student_roll, double Student_fee) {
        this.Student_name = Student_name;
        this.Student_roll = Student_roll;
        this.Student_fee = Student_fee;
    }
    void Display() {
        System.out.println("Name: " + Student_name + "\nRoll: " + Student_roll + "\nFee: " + Student_fee);
    }
}

public class Lab8_Q2 {
    public static void main(String[] args) {
        Student s1 = new Student("Suraj", 101, 10000);
        s1.Display();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 8 % java Lab8_Q2
Name: Suraj
Roll: 101
Fee: 10000.0
```

```

Lab8_Q3.java

//Lab 8: Question 3
class Student {
    String Student_name;
    int Student_roll;
    double Student_fee;
    Student(String Student_name, int Student_roll, double Student_fee) {
        this.Student_name = Student_name;
        this.Student_roll = Student_roll;
        this.Student_fee = Student_fee;
    }
    Student() {
        this("Gautam", 102, 50000);
    }
    void Show(Student obj) {
        this.Display();
        System.out.println("Displaying From Show Function");
    }
    void Display() {
        System.out.println("Name: " + Student_name + "\nRoll: " + Student_roll + "\nFee: " + Student_fee);
    }
}

public class Lab8_Q3 {
    public static void main(String[] args) {
        Student s1 = new Student("Suraj", 101, 10000);
        Student s2 = new Student();
        s2.Show(s1);
        s2.Show(s2);
    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 8 % java Lab8_Q3
Name: Gautam
Roll: 102
Fee: 50000.0
Displaying From Show Function
Name: Gautam
Roll: 102
Fee: 50000.0
Displaying From Show Function
itzspgautam@Surajs-MacBook-Pro 8 %

```

OUTPUT

Lab Experiment 9

(Enum, Exception handling, Inheritance, Method Overriding, super keyword)

EXP 1: To demonstrate the use of enum in Java by creating a simple program that displays a message based on the day of the week. The program prompts the user to enter a day, matches the input with the corresponding enum constant, and prints a message indicating whether it is a weekday or a weekend. If the user enters an invalid day name, the program handles it by throwing an `IllegalArgumentException` and prints "Invalid Day".

EXP 2: Illustrate the execution of constructors in multi-level inheritance with three Java classes – `plate(length, width)`, `box(length, width, height)`, `woodbox (length, width, height, thick)`. Print all the information (length, width, height, thick) from the `WoodBox` class.

EXP 3. To write a Java program that demonstrates inheritance by calculating rent for different types of vehicles — Car, Bike, and Bus. The program should perform the following operations:

- Allow the user to select a vehicle and enter the number of days for which they want to rent it.
- Finally, calculate and display the total rent amount for the selected vehicle and rental duration.

(Optional)

EXP 4. Write a Java program to implement a Vehicle Rental System using inheritance to calculate the rent for different types of vehicles such as Car, Bike, and Bus, following the guidelines below:

The base class `Vehicle` should contain common attributes such as:

- Vehicle Number
- Vehicle Type
- Model Name
- Rental Rate per Day

The derived classes `Car`, `Bike`, and `Bus` should extend the `Vehicle` class and override the `calculateRent()` method to provide customized rent calculation logic specific to each vehicle type.

The program should perform the following operations:

- Prompt the user to enter the required vehicle type (e.g., car, bike, or bus).
- Based on the user's input, display the available vehicles of that type, including details such as model name, seat capacity, and rental rate per day.
- Allow the user to select a vehicle and enter the number of days for which they want to rent it.
- Finally, calculate and display the total rent amount for the selected vehicle and rental duration.

```

Lab9_Q1.java

//Lab 9: Question 1
import java.util.Scanner;
enum Day {MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY}
public class Lab9_Q1{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a day:");
        String input = sc.nextLine().toUpperCase();

        try{
            Day day = Day.valueOf(input);
            if(day == Day.SUNDAY || day == Day.SATURDAY){
                System.out.println("Its Weekend. Enjoy!");
            }
            else{
                System.out.println("Its Weekday. :/");
            }
        }catch(IllegalArgumentException e){
            System.out.println("Error: Invalid day");
        }
    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q1
Enter a day:
Monday
Its Weekday. :/

```

OUTPUT

```

Lab9_Q2.java

//Lab 9: Question 2
class Plate {
    int length, width;
    Plate(int length, int width) {
        this.length = length; this.width = width;
        System.out.println("Plate Information(length and width) assigned.");
    }
}
class Box extends Plate {
    int height;
    Box(int length, int width, int height) {
        super(length, width); this.height = height;
        System.out.println("Box Information (height) assigned.");
    }
}
class WoodBox extends Box {
    int thick;
    WoodBox(int length, int width, int height, int thick) {
        super(length, width, height); this.thick = thick;
        System.out.println("WoodBox Information(thick) assigned.");
    }
    void display() {
        System.out.println("=====ASSIGNED INFO=====");
        System.out.println("Length = " + length + "\n" + "Width = " + width + "\n" + "Height = " + height + "\n" + "Thick = " + thick);
    }
}
public class Lab9_Q2 {
    public static void main(String[] args) {
        WoodBox wb = new WoodBox(10, 5, 8, 2);
        wb.display();
    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q1
Enter a day:
Monday
Its Weekday. :/
itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q1
Enter a day:
Sunday
Its Weekend. Enjoy!
itzspgautam@Surajs-MacBook-Pro 9 % 

```

OUTPUT

```

Lab9_Q3.java

//Lab 9: Question 3
import java.util.Scanner;
class Vehicle {
    int rentPerDay;
    void calculateRent(int day) {
        System.out.println("Total Rent= " + rentPerDay * day + "INR (" + day + " Days X " + rentPerDay + " INR)");
    }
}
class Bike extends Vehicle {
    Bike() {
        rentPerDay = 1000;
        System.out.println("Rent per day of Bike= " + rentPerDay);
    }
}
class Car extends Vehicle {
    Car() {
        rentPerDay = 4000;
        System.out.println("Rent per day of Car= " + rentPerDay);
    }
}
class Bus extends Vehicle {
    Bus() {
        rentPerDay = 6000;
        System.out.println("Rent per day of Bus= " + rentPerDay);
    }
}
public class Lab9_Q3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("=====SELECT VEHICLE TO RENT=====");
        System.out.println("1.Bike\n2.Car\n3.Bus");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();

        Vehicle v = null;
        switch (choice) {
            case 1: v = new Bike(); break;
            case 2: v = new Car(); break;
            case 3: v = new Bus(); break;
            default: System.out.println("Invalid Choice! closing the program."); return;
        }
        System.out.print("Enter days to rent: ");
        int day = sc.nextInt();
        v.calculateRent(day);
    }
}

```

CODE

```

itzspgautam@Surajs-MacBook-Pro 9 % javac Lab9_Q3.java
itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q3
=====SELECT VEHICLE TO RENT=====
1.Bike
2.Car
3.Bus
Enter choice: 1
Rent per day of Bike= 1000
Enter days to rent: 10
Total Rent= 10000INR (10 Days X 1000 INR)
itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q3
=====SELECT VEHICLE TO RENT=====
1.Bike
2.Car
3.Bus
Enter choice: 3
Rent per day of Bus= 6000
Enter days to rent: 10
Total Rent= 60000INR (10 Days X 6000 INR)
itzspgautam@Surajs-MacBook-Pro 9 % █

```

OUTPUT

Lab9_Q4_Optional.java

CODE

```
//Lab 9: Question 4 (Optional)
import java.util.Scanner;
class Vehicle {
    String number, type, model;
    double rate;
    Vehicle(String number, String type, String model, double rate) {
        this.number = number;
        this.type = type;
        this.model = model;
        this.rate = rate;
    }
    double calculateRent(int days) {
        return rate * days;
    }
}
class Car extends Vehicle {
    Car() { super("C101", "Car", "Honda City", 1800); }
    double calculateRent(int d) { return super.calculateRent(d) + 500; }
}
class Bike extends Vehicle {
    Bike() { super("B201", "Bike", "Royal Enfield", 700); }
}
class Bus extends Vehicle {
    Bus() { super("BS301", "Bus", "Volvo AC", 2500); }
    double calculateRent(int d) { return super.calculateRent(d) + 1000; }
}
public class Lab9_Q4_Optional {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter vehicle type (car/bike/bus): ");
        String type = sc.next().toLowerCase();

        Vehicle v;
        if (type.equals("car")) v = new Car();
        else if (type.equals("bike")) v = new Bike();
        else if (type.equals("bus")) v = new Bus();
        else { System.out.println("Invalid type!"); return; }

        System.out.print("Enter number of days: ");
        int days = sc.nextInt();

        double rent = v.calculateRent(days);
        System.out.println("\nVehicle: " + v.model + "\nDays: " + days + "\nTotal Rent: ₹" + rent);
        sc.close();
    }
}
```

```
● itzspgautam@Surajs-MacBook-Pro 9 % javac Lab9_Q4_Optional.java
● itzspgautam@Surajs-MacBook-Pro 9 % java Lab9_Q4_Optional
Enter vehicle type (car/bike/bus): car
Enter number of days: 10

Vehicle: Honda City
Days: 10
Total Rent: ₹18500.0
● itzspgautam@Surajs-MacBook-Pro 9 %
```

OUTPUT



Thank
You

SURAJ PRAKASH GAUTAM

